YUKTHA

Deepfake Defense System

Multi-modal AI for authentic digital content verification

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Guide - Mrs. V Manjula

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Agenda

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Abstract

Deepfake media generated using advanced generative models poses a serious threat to digital trust, enabling misinformation, identity fraud, and social manipulation. Existing deepfake detection approaches are largely model-centric, focusing on either audio or video modalities in isolation, often relying on naive fusion strategies and lacking explainability and deployment readiness. In this work, we propose a lean, multimodal deepfake detection system that integrates spatial, temporal, audio, frequency, physiological, and lip-sync cues through a learning-based decision-level fusion framework. The proposed system processes audio and video using independent pipelines and performs confidence-aware fusion to improve robustness against partial and fully synthetic manipulations. Lightweight audio–visual lip-sync verification is employed by default, with optional escalation to advanced phoneme-level models such as SyncNet or AV-HuBERT under low-confidence, talking-head scenarios. Additionally, the system provides explainable outputs using visual and audio heatmaps, temporal plots, and confidence breakdowns, enabling forensic interpretability. Designed with real-world deployment in mind, the system adopts an API-driven architecture suitable for web, mobile, and enterprise use. This work bridges the gap between academic deepfake detection models and practical, trustworthy detection systems.

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Summary of Literature Review

Video Deepfake Detection Approaches

State-of-the-Art Architectures:

- **XceptionNet:** Achieves 95% accuracy on FaceForensics++ through frequency domain analysis
- **MesoNet (MesoInception-4):** 84% on DeepFake TIMIT with lightweight design
- **Capsule Networks:** 96.6% accuracy by capturing spatial inconsistencies
- **ResNeXt-50 + LSTM:** 94.3% on Celeb-DF v2 (hybrid spatial-temporal approach)
- **EfficientNet-B4 + LSTM:** 93.8% on DFDC dataset

Critical Insight: CNN-only approaches rely heavily on global frame features and often miss localized manipulation artifacts. The most robust detectors combine:

- Spatial analysis (CNN extracting pixel-level inconsistencies)
- Temporal analysis (LSTM/RNN modeling frame-to-frame dynamics)
- Attention mechanisms (guiding focus to artifact-prone regions)

Audio Deepfake Detection Approaches

Feature Extraction Methods:

- **MFCC (Mel-Frequency Cepstral Coefficients):** Mimics human auditory perception; achieved 98.83% accuracy with SVM on clean datasets
- **Mel-Spectrograms:** Better captures frequency variations; used with VGG-16 for 93% accuracy on Fake-or-Real dataset
- **Spectrogram Analysis:** Reveals temporal inconsistencies in synthetic voice generation

Effective Architectures:

- **CNN-RNN Hybrid:** CNN extracts spectral features from spectrograms; RNN/LSTM models temporal sequences
- **BiLSTM Networks:** Bidirectional temporal analysis captures both forward and backward dependencies
- **Attention-Based Models:** Focus on frequency bands most susceptible to manipulation

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Summary of Literature Review

Multi-Modal Deepfake Detection

Fusion Strategy	Description	Advantage	Use Case
Simple, modular; easy interpretation	Independent classifiers for each modality; decisions averaged at output	Late Fusion	Lightweight systems
Balanced speed/accuracy	Features concatenated before final classification layer	Mid Fusion	Standard deployments
Superior performance; captures cross-modal patterns	Raw features concatenated early; transformer processes combined representation	Early Fusion	High-precision applications

Key Finding: Early Fusion with Transformer architectures outperforms late fusion by 3-5% on multimodal datasets (FakeAVCeleb, DDL-AV)

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Proposed System Architecture

🔴 Gap 1: Lack of True Multi-Modal Fusion

Gap:

Most works use only audio or video. Even audio-visual methods rely on feature concatenation or simple averaging, with no learning-based trust between modalities.

Our Approach:

Independent audio and video pipelines with decision-level, learning-based fusion and confidence calibration for adaptive modality weighting.

🔴 Gap 2: Weak Lip-Sync Analysis

Gap:

Lip-sync and phoneme-lip alignment are largely ignored, especially for talking-head deepfakes.

Our Approach:

Lightweight lip motion vs audio energy correlation, with optional SyncNet / AV-HuBERT for low-confidence cases.

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🔴 Gap 3: Audio and Video Treated Separately

Gap:

Audio spoofing and video manipulation are studied in isolation, ignoring cross-modal inconsistencies.

Our Approach:

Explicit audio–video cross-verification through joint score fusion, enabling detection of partial manipulations.

🔴 Gap 4: Lack of Explainability

Gap:

Most methods report accuracy only, offering black-box predictions.

Our Approach:

Built-in explainability using Grad-CAM, audio heatmaps, and temporal visualizations, delivered as a forensic report.

🔴 Gap 5: No Deployment Focus

Gap:

Existing works remain offline academic prototypes with no real-world integration.

Our Approach:

API-driven, modular system designed for web, mobile, and enterprise deployment.

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System Requirements

1. Hardware Requirements Development / Training Environment

- GPU: NVIDIA T4 / RTX 3060 or higher (minimum 8 GB VRAM; recommended 16 GB+)
- CPU: Quad-core Intel i5 / Ryzen 5 or higher
- RAM: Minimum 16 GB (Recommended: 32 GB for multi-modal processing)
- Storage:
  - Minimum: 200 GB free space
  - Recommended: 1 TB SSD (for datasets + features)
- Internet: Stable broadband connection for model downloads

Deployment Environment

- CPU server or GPU server (depending on inference speed requirement)
- 8 GB RAM minimum
- SSD storage for fast media access
- Load balancer for production use

2. Software Requirements

Operating System Windows 10/11 (development only)

Programming Environment

- **Python 3.9+**
- **CUDA Toolkit 11+**
- **cuDNN compatible with CUDA version**

Libraries & Frameworks

Core AI

- **PyTorch**
- **TorchVision**
- **timm (EfficientNet)**
- **Transformers (Wav2Vec2 / AV-HuBERT)**

Computer Vision

- **OpenCV**
- **MTCNN / InsightFace**
- **NumPy**

Audio Processing: Librosa, SciPy

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System Requirements

3. Model Requirements

The system requires pretrained or trained versions of:

- EfficientNet-B4 (spatial branch)
- Temporal Convolutional Network
- Audio CNN + TCN
- Lip-sync model (SyncNet or AV-HuBERT)
- Fusion MLP model

All models must be stored in serialized checkpoint format (.pth / .pt).

4. Data Requirements

- Supported Formats:
  - Video: MP4, AVI, MOV
  - Audio: WAV, MP3
- Input Constraints:
  - Minimum resolution: 224×224
  - Minimum duration: 1 second, Maximum duration: configurable
- Dataset Requirements:
  - Labeled real vs fake media
  - Balanced class distribution preferred

5. Performance Requirements

- Frame processing latency: < 50 ms per frame
- Full video inference time:
  - GPU: < 5 seconds (short clips)
  - CPU: < 20 seconds
- Accuracy target:
  - AUC ≥ 0.90 on benchmark dataset
- Throughput:
  - ≥ 10 videos/minute (GPU inference server)

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System Requirements

6. Functional Requirements

The system must:

- Accept video/audio inputs
- Extract frames and audio signals
- Detect faces
- Compute spatial, temporal, audio, and physiological features
- Perform multi-modal fusion
- Output authenticity score
- Generate explainability report

7. Non-Functional Requirements Reliability

- Must handle corrupted files gracefully
- Must resume processing after interruption

Scalability

- Should support horizontal scaling via container orchestration

Security

- Must sanitize uploaded files
- Must prevent malicious media execution

Usability

- API must return structured JSON results
- Interface must show confidence and explanation

8. Deployment Requirements

Production deployment must support:

- Docker containers
- Kubernetes orchestration
- Auto-scaling
- Logging & monitoring
- Model versioning

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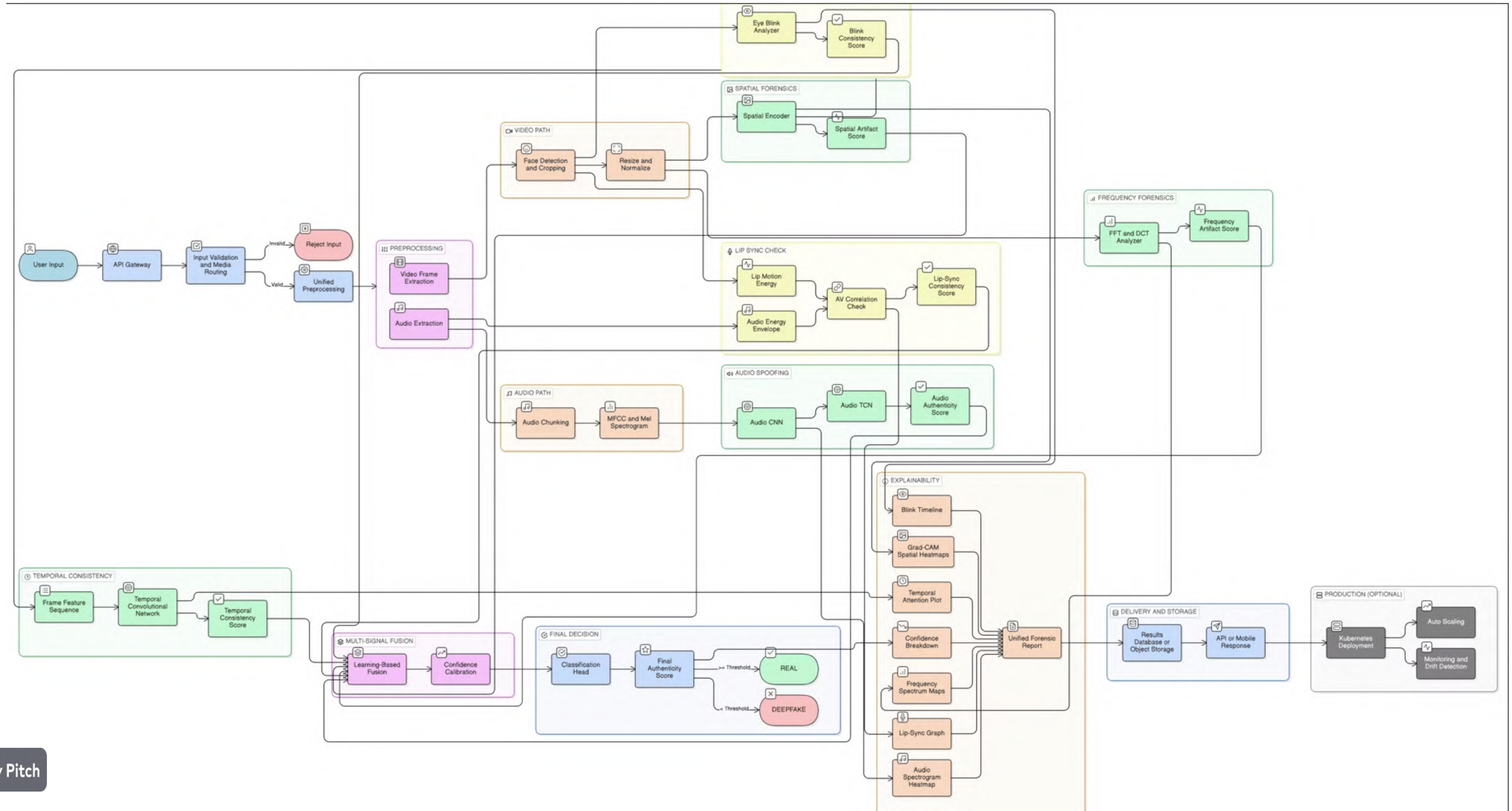
Architecture Design



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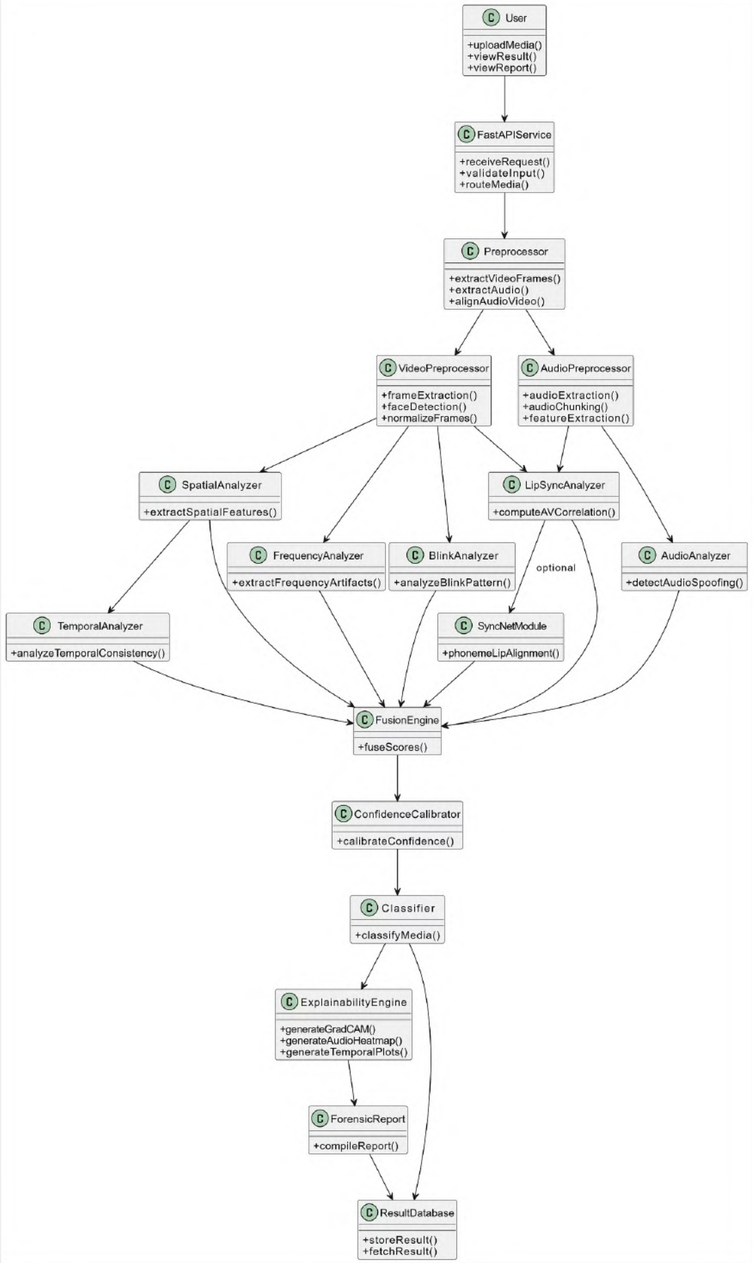

UML Diagrams - Class Diagram

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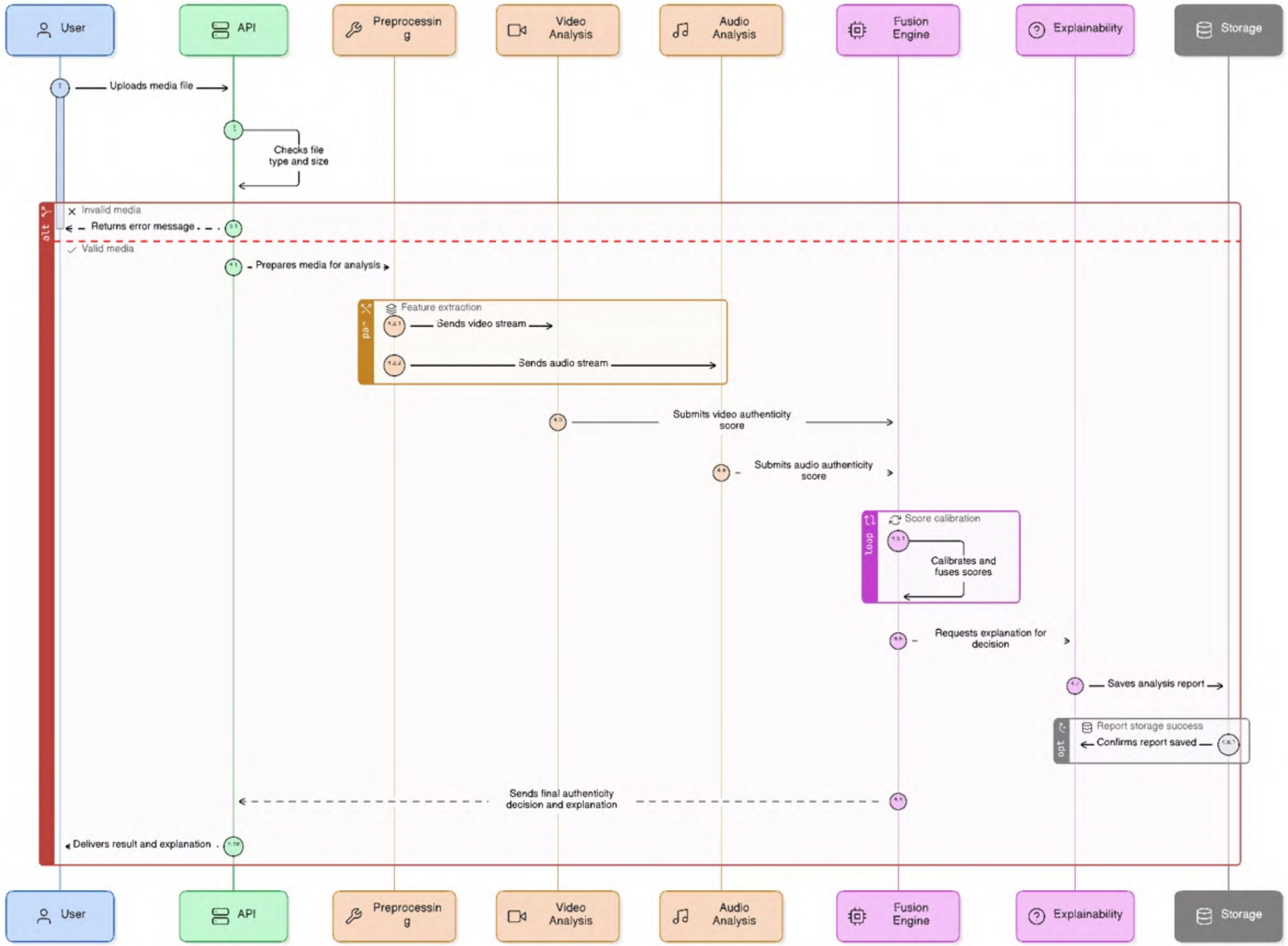
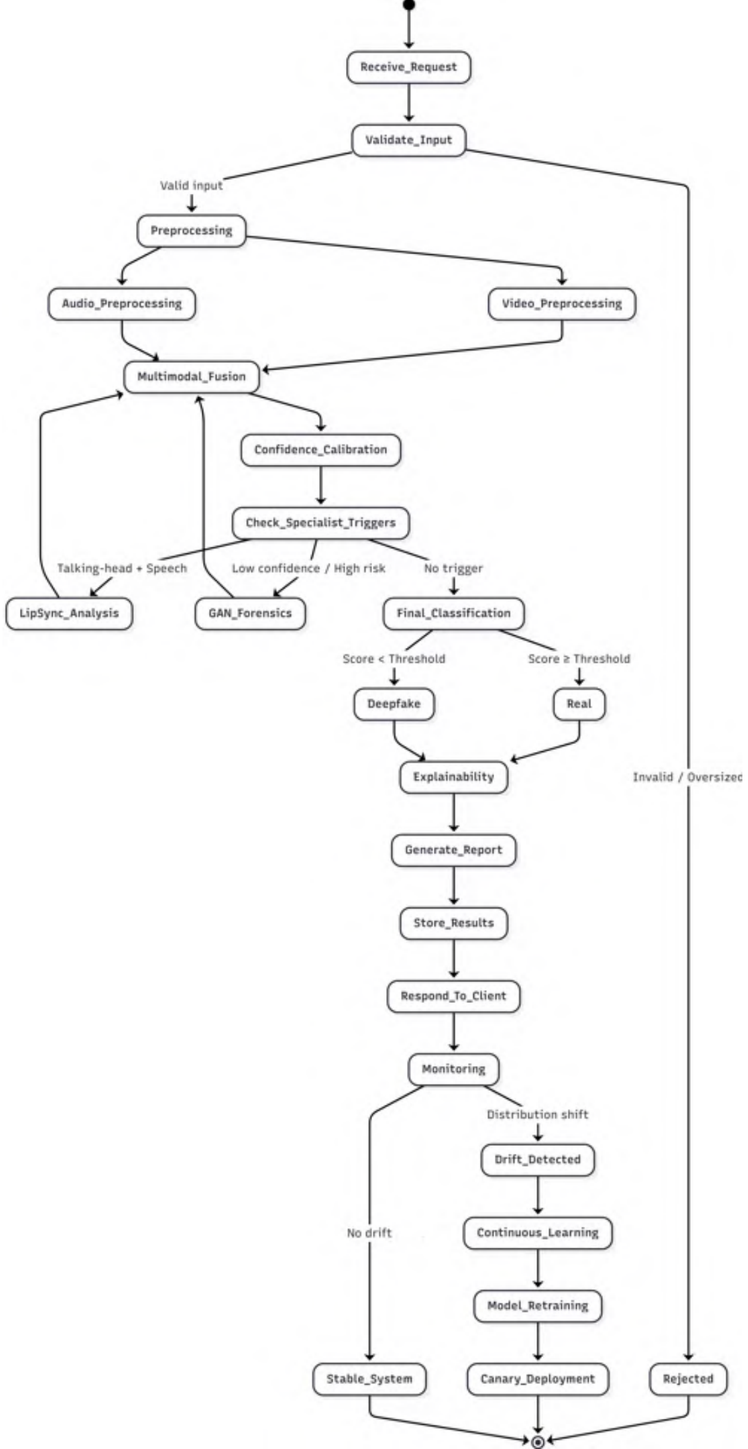
UML Diagrams - Sequence Diagram



Try Pitch



UML Diagrams - State Diagram

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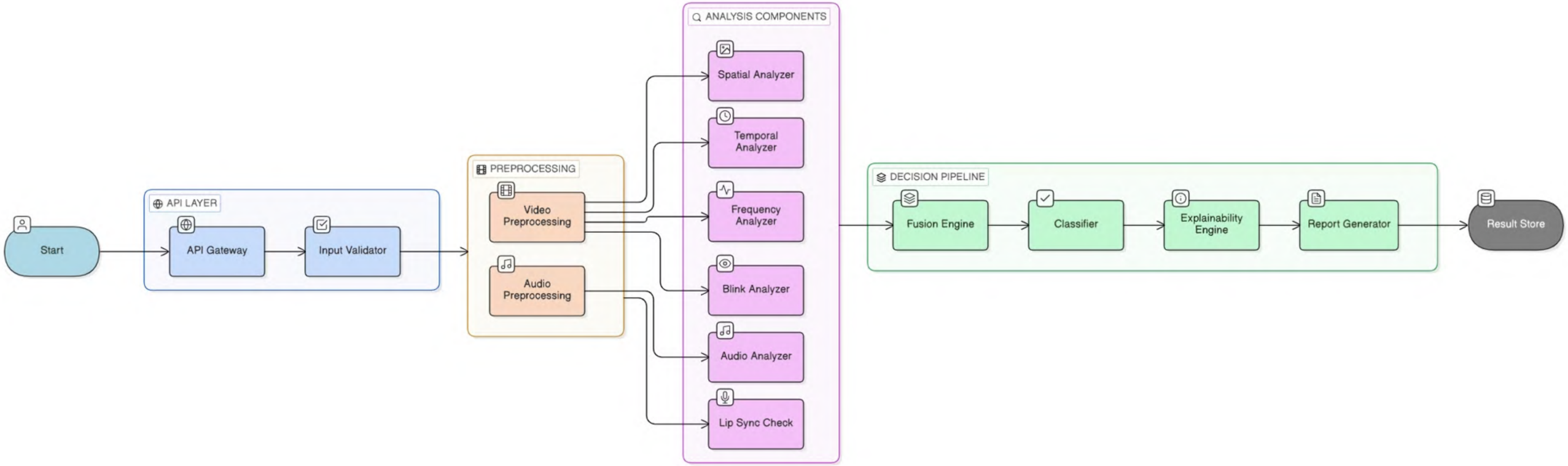
UML Diagrams - Component Diagram



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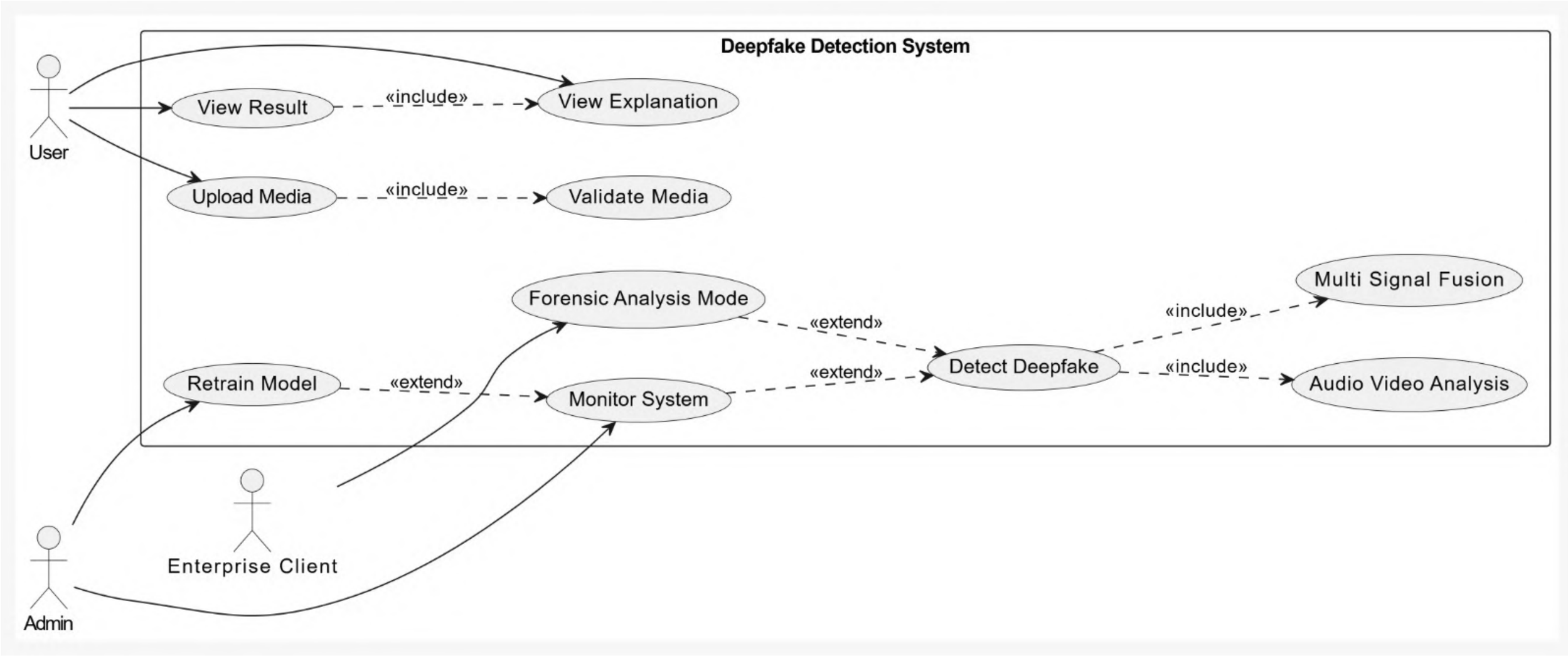
UML Diagrams - Use Case Diagram



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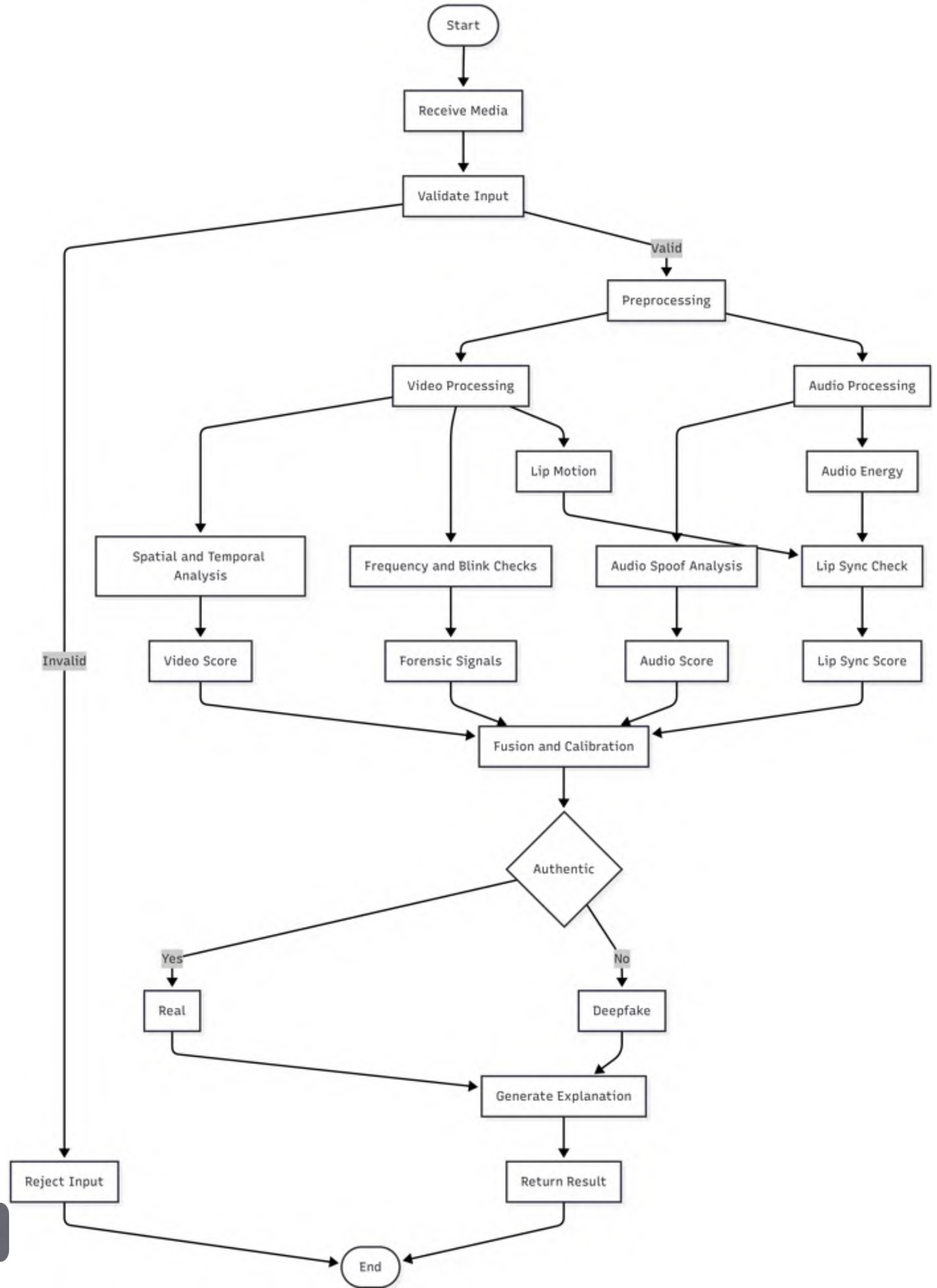

UML Diagrams - Activity Diagram

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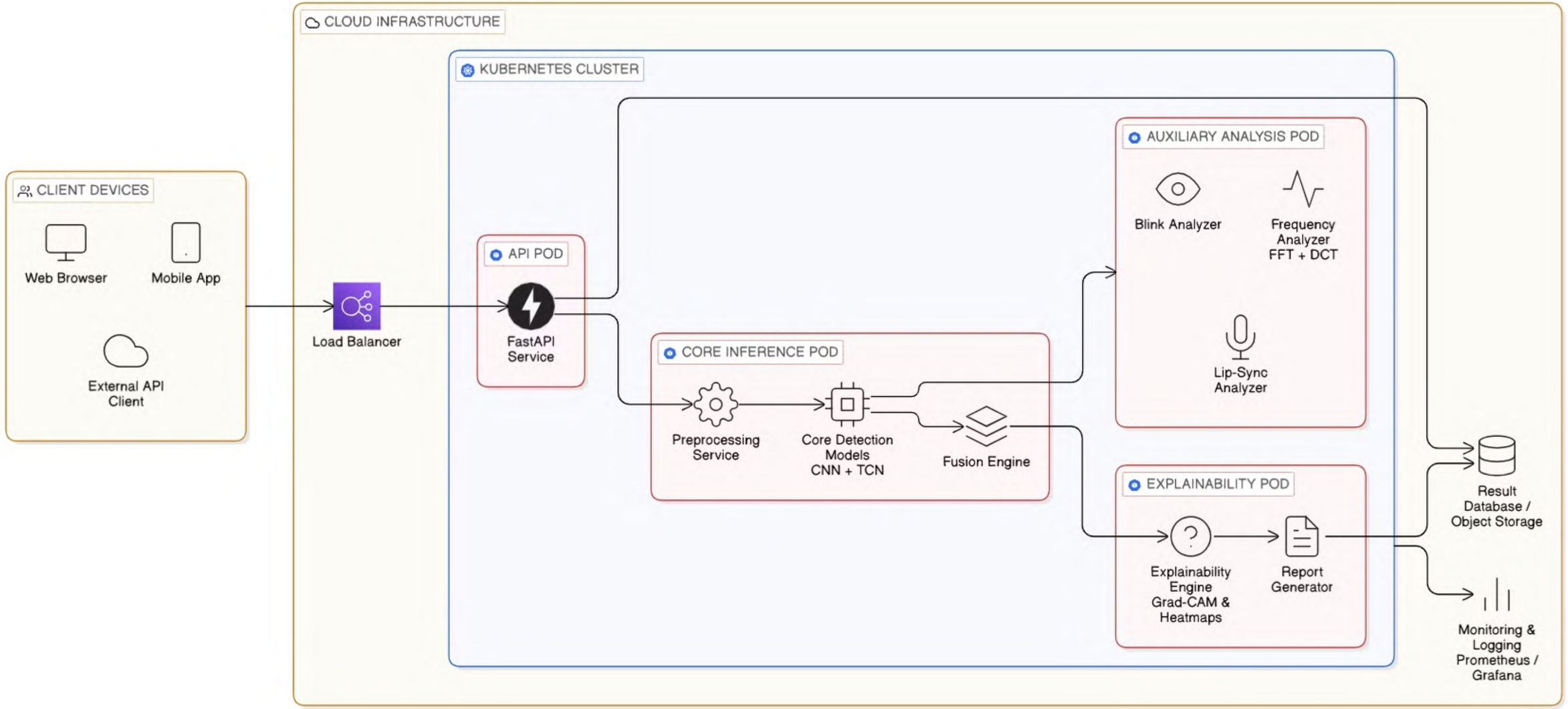
UML Diagrams - Deployment Diagram



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Core Modules

Module 1: Input Handler & Validation Layer

Purpose: Handle and validate incoming media inputs from multiple sources.

Responsibilities:

- Accept video/audio uploads
- Detect file format and type
- Validate size, duration, codec
- Route media to correct pipeline
- Reject corrupted or unsupported inputs

Technologies: FastAPI, Python MIME, FFmpeg, OpenCV

✅ **Module 2: Unified Preprocessing Engine**

Purpose: Prepare multimedia data for analysis across all forensic branches.

Responsibilities:

- Video frame extraction (15–30 FPS)
- Audio extraction
- Frame normalization (224×224)
- Data buffering for batch processing,Metadata extraction

Technologies: OpenCV, FFmpeg, NumPy, PIL

✅ **Module 3: Face Processing Module**

Purpose: Detect and isolate facial regions for forensic analysis.

Responsibilities:

- Face detection (MTCNN)
- Face cropping
- Landmark detection
- Face alignment
- Region tracking across frames

Technologies: MTCNN, Dlib, OpenCV

✅ **Module 4: Spatial Forensics Module (Core)**

Purpose: Detect visual manipulation artifacts in facial frames.

Responsibilities:

- Extract spatial features
- Detect blending artifacts
- Identify texture inconsistencies
- Compute spatial authenticity score

Model: EfficientNet-B4 CNN

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Core Modules

Module 5: Frequency Forensics Module

Purpose: Identify manipulation artifacts invisible in pixel space.

Responsibilities:

- FFT transformation
- DCT analysis
- Frequency band anomaly detection
- Compression artifact analysis

Technologies: NumPy FFT, SciPy DCT

✅ **Module 6: Temporal Consistency Module (Core)**

Purpose: Detect unnatural motion patterns across frames.

Responsibilities:

- Build frame feature sequence
- Capture motion consistency
- Detect frame transition anomalies
- Model long-range temporal patterns

Model: Temporal Convolutional Network (TCN)

✅ **Module 7: Audio Authenticity Module (Core)**

Purpose: Detect synthetic or manipulated speech.

Responsibilities:

- Extract MFCC and Mel spectrogram
- Analyze speech spectral structure
- Model temporal speech patterns
- Detect vocoder artifacts

Model: CNN + Audio TCN

✅ **Module 8: Physiological Signal Analyzer**

Purpose: Validate biological realism of subject.

Responsibilities:

- Blink detection
- Blink frequency analysis
- Eye aspect ratio computation
- Heart-rate (rPPG) signal estimation
- Physiological consistency scoring

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Core Modules

Module 9: Lip-Sync Verification Module Lightweight Layer (Default)

Purpose: Quickly verify audio–visual consistency.

Responsibilities:

- Lip motion energy extraction
- Audio energy envelope extraction
- Correlation computation

Advanced Layer (Triggered Mode)

Purpose: Perform precise phoneme–lip alignment verification.

Trigger Conditions:

- Low lightweight score
- Talking head detected
- High-risk mode enabled

Models: SyncNet / AV-HuBERT

✅ **Module 10: Multi-Signal Fusion Engine**

Purpose: Combine evidence from all modalities into a unified decision.

Responsibilities:

- Aggregate modality scores
- Learn modality importance weights
- Handle missing modalities
- Produce unified authenticity logit

Models: MLP Fusion Network / XGBoost Ensemble

Module 11: Confidence Calibration & Decision Module

Purpose: Generate reliable probability estimates and final classification.

Responsibilities:

- Probability calibration
- Threshold decision logic
- Risk-aware decision adjustment
- Output final authenticity score

Methods: Temperature scaling, logistic calibration

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Core Modules

Module 12: Explainability & Forensic Report Engine

Purpose: Provide interpretable evidence for predictions.

Responsibilities:

- Spatial heatmaps (Grad-CAM)
- Frequency maps
- Temporal attention visualization
- Spectrogram heatmaps
- Blink timeline plots
- Lip-sync graphs
- Confidence breakdown charts

Output:

Unified forensic report

Module 13: Output Delivery Layer

Purpose: Deliver results to client systems.

Responsibilities:

- JSON API response
- Result storage
- Evidence linking
- Dashboard visualization

Technologies: FastAPI, Object Storage, Database

Module 14: Deployment & Monitoring Layer

Purpose: Enable scalable, production-ready deployment.

Responsibilities:

- Container orchestration
- Auto scaling
- Performance monitoring
- Drift detection
- Logging and alerting

Technologies: Kubernetes, Prometheus, Grafana

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Implementation

Implemented a multi-modal deepfake detection pipeline that preprocesses DFDC videos to extract spatial, motion, audio, and physiological features using GPU-accelerated models. Each modality is modeled with dedicated neural networks and trained separately for authenticity prediction. The learned embeddings are combined using a late-fusion MLP to produce the final deepfake classification. The system includes checkpointing, mixed-precision training, and robust data processing for efficient large-scale experimentation.

```
[2026-02-10 08:36:53] face Epoch 1 finished in 37.0s loss 0.3461 train_auc 0.5528
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:37:00] Eval face: AUC 0.5691
[2026-02-10 08:37:00] face checkpoint saved to /content/drive/MyDrive/dfdc_train_ckpts/face_best.pth
/tmp/ipython-input-602814789.py:293: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:37:36] face Epoch 2 finished in 36.1s loss 0.2086 train_auc 0.7616
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:37:43] Eval face: AUC 0.7470
[2026-02-10 08:37:43] face checkpoint saved to /content/drive/MyDrive/dfdc_train_ckpts/face_best.pth
/tmp/ipython-input-602814789.py:293: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
```

```
[2026-02-10 08:39:54] Eval face: AUC 0.9250
[2026-02-10 08:39:54] face checkpoint saved to /content/drive/MyDrive/dfdc_train_ckpts/face_best.pth
/tmp/ipython-input-602814789.py:293: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:40:31] face Epoch 6 finished in 37.1s loss 0.1230 train_auc 0.9546
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:40:38] Eval face: AUC 0.9284
[2026-02-10 08:40:38] face checkpoint saved to /content/drive/MyDrive/dfdc_train_ckpts/face_best.pth
[2026-02-10 08:40:38] face best val AUC = 0.9284
[2026-02-10 08:40:38] Training Motion encoder ...
/tmp/ipython-input-602814789.py:293: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
```

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Implementation

```
[2026-02-10 08:41:15] motion Epoch 1 finished in 36.6s loss 0.3968 train_auc 0.5255
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:41:22] Eval motion: AUC 0.4918
[2026-02-10 08:41:22] motion checkpoint saved to /content/drive/MyDrive/dfdc_train_ckpts/motion_best.pth
/tmp/ipython-input-602814789.py:293: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:41:58] motion Epoch 2 finished in 36.3s loss 0.2923 train_auc 0.5340
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
```

```
[2026-02-10 08:44:11] Eval motion: AUC 0.4560
/tmp/ipython-input-602814789.py:293: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:44:47] motion Epoch 6 finished in 35.5s loss 0.2824 train_auc 0.5495
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:44:54] Eval motion: AUC 0.4826
[2026-02-10 08:44:54] motion best val AUC = 0.4918
```

```
[2026-02-10 08:44:54] Training Audio encoder ...
/tmp/ipython-input-602814789.py:293: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:45:29] audio Epoch 1 finished in 35.2s loss 0.4079 train_auc 0.5021
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:45:36] Eval audio: AUC 0.4386
[2026-02-10 08:45:36] audio checkpoint saved to /content/drive/MyDrive/dfdc_train_ckpts/audio_best.pth
/tmp/ipython-input-602814789.py:293: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:46:12] audio Epoch 2 finished in 36.3s loss 0.2442 train_auc 0.5582
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
```

```
[2026-02-10 08:49:04] audio Epoch 6 finished in 35.8s loss 0.2313 train_auc 0.6463
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:49:11] Eval audio: AUC 0.3250
[2026-02-10 08:49:11] audio best val AUC = 0.4386
```

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Implementation

```
[2026-02-10 08:49:11] Training Phys encoder ...
/tmp/ipython-input-602814789.py:293: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:49:47] phys Epoch 1 finished in 36.0s loss 0.5587 train_auc 0.5092
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:49:53] Eval phys: AUC 0.4949
[2026-02-10 08:49:53] phys checkpoint saved to /content/drive/MyDrive/dfdc_train_ckpts/phys_best.pth
/tmp/ipython-input-602814789.py:293: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:50:29] phys Epoch 2 finished in 36.1s loss 0.3778 train_auc 0.5003
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
```

```
[2026-02-10 08:53:21] phys Epoch 6 finished in 35.7s loss 0.2833 train_auc 0.5315
/tmp/ipython-input-602814789.py:331: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:53:27] Eval phys: AUC 0.4580
[2026-02-10 08:53:27] phys best val AUC = 0.4949
[2026-02-10 08:53:27] Per-stream training complete. Results so far: {'face': np.float64(0.9284442285899304), 'motion': np.float64(0.49178403755868544), 'audio': np.float64(0.43
[2026-02-10 08:53:27] Loaded face weights from /content/drive/MyDrive/dfdc_train_ckpts/face_best.pth
[2026-02-10 08:53:27] Loaded motion weights from /content/drive/MyDrive/dfdc_train_ckpts/motion_best.pth
[2026-02-10 08:53:27] Loaded audio weights from /content/drive/MyDrive/dfdc_train_ckpts/audio_best.pth
[2026-02-10 08:53:27] Loaded phys weights from /content/drive/MyDrive/dfdc_train_ckpts/phys_best.pth
[2026-02-10 08:53:27] Fusion model created. Training fusion head only.
/tmp/ipython-input-602814789.py:440: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated. Please use `torch.amp.GradScaler('cuda', args...)` instead.
  scaler_f = torch.cuda.amp.GradScaler(enabled=USE_AMP)
/tmp/ipython-input-602814789.py:449: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:54:10] Fusion Epoch 1 done in 36.3s  val_auc 0.8540
```

```
[2026-02-10 08:57:02] Saved fusion_best.pth
/tmp/ipython-input-602814789.py:449: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
  with torch.cuda.amp.autocast(enabled=USE_AMP):
[2026-02-10 08:57:44] Fusion Epoch 6 done in 36.5s  val_auc 0.9635
[2026-02-10 08:57:44] Saved fusion_best.pth
[2026-02-10 08:57:44] FINAL RESULTS (AUC):
Face only       0.9284
Motion only     0.4918
Audio only      0.4386
Physiology only 0.4949
Fusion (all)    0.9635
[2026-02-10 08:57:44] Training complete. Checkpoints and results saved to: /content/drive/MyDrive/dfdc_train_ckpts
```

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Current Project Status

- ☐ Input Handler & Validation Layer
- ☑ Unified Preprocessing Engine
- ☑ Face Processing Module
- ☑ Spatial Forensics Module
- ☐ Frequency Forensics Module
- ☑ Temporal Consistency Module
- ☑ Audio Authenticity Module
- ☑ Physiological Signal Analyzer
- ☐ Lip-Sync Verification Module
- ☑ Multi-Signal Fusion Engine
- ☐ Confidence Calibration & Decision Module
- ☐ Explainability & Forensic Report Engine
- ☐ Output Delivery Layer
- ☐ Deployment & Monitoring Layer

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Thank You

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Guide - Mrs. V Manjula

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